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State of the Art

The Cystic Fibrosis Gender Gap: Potential Roles of Estrogen

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Summary. Cystic fibrosis (CF) is a complex, multi-system, autosomal recessive disease predominantly affecting Caucasians that leads to vigorous airway inflammation and chronic respiratory infection, commonly with *Pseudomonas aeruginosa*. A variety of factors significantly modify the progression and severity of CF lung disease and the timing of the resulting mortality. We summarize here data indicating that there is in CF a female disadvantage in survival and morbidity, called the "CF gender gap". Although controversy exists regarding the nature and relative importance of the various contributing mechanisms involved, gender affects the progression of CF disease with respect to lung infection, decline in pulmonary function and nutritional status. These interrelated factors in turn have a negative impact on survival. This review will emphasize the increasing evidence that suggest a role for the effects of gender, and particularly the female sex hormone estrogen, on infection, inflammation and transepithelial ion transport, all major determinants of CF lung disease. Future elucidation of the pathophysiology of hormonal aggravation of CF lung disease may pave the way for novel therapeutic interventions. This, combined with the magnitude of the gender gap in CF mortality, strongly suggests that further work in this field is well justified. **Pediatr Pulmonol. 2014; 49:309–317.** © 2013 Wiley Periodicals, Inc.

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INTRODUCTION

Cystic fibrosis (CF) is a complex, multi-system, autosomal recessive disease predominantly affecting Caucasians that leads to vigorous airway inflammation and chronic respiratory infection, commonly with *Pseudomonas aeruginosa*. These processes cause severe lung damage, the principal cause of death.[1–3] A variety of environmental, anatomical, physiological, cultural, and genetic factors, both known and as yet to be identified, significantly modify the progression and severity of CF lung disease and the timing of the resulting mortality.[4] We summarize here data indicating that there is in CF a female disadvantage in survival and morbidity, called the "CF gender gap." Although controversy exists regarding the nature and relative importance of the various contributing mechanisms involved, this review will emphasize the increasing evidence that suggests a role

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A systematic review was conducted using PubMed. Terms included "cystic fibrosis," "gender gap," "sex," "survival," "mortality," "inflammation," "estrogen," and "estradiol." Searches were restricted to the English language with no age limit. Reference lists and conference proceedings were also reviewed.

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for the effects of gender, and particularly the female sex hormone estrogen, on infection, inflammation and trans-epithelial ion transport, all major determinants of CF lung disease.

METHODS

Systematic searches of the literature were conducted in PubMed using combinations of keywords, nested in the following patterns: (cystic fibrosis) AND (gender gap); ((cystic fibrosis) AND (gender gap)) AND mortality; ((cystic fibrosis) AND (gender gap)) AND estrogen. Each search was then repeated, either substituting the term "sex" or "gender" for "gender gap" and/or "estradiol" for "estrogen," and then restricted by adding the term "inflammation."

GENDER—EFFECT ON CF MORTALITY/SURVIVAL

CF clinicians have long suspected a CF gender gap and strong supporting evidence has been available for decades. A series of studies from the late 1990s found that among CF patients, females die younger than males,[5,6] although Rosenfeld et al.[5] reported that females were more likely to die than males in the age group 1–20 years whereas outside this age range survival rates were not significantly different. More recently, large data sets from well-established national CF patient registries with good (~85%+) coverage of their respective communities document a gender-based difference in age of death and/or in predicted median survival. At Canadian CF clinics, with 3,913 patients registered in 2011, median predicted survival ages are estimated to be 50.6 years for males compared to 43.2 years for females (Canadian CF Registry 2011 Annual Report; Toronto, 2013; http://www.cysticfibrosis.ca/assets/files/pdf/Registry2011FINALOnlineEN.pdf). As illustrated in Figure 1, Canadian data from 1974 to 2010 consistently showed this gender gap in median survival.

As of 2011, males accounted for 52.7% of all Canadians living with CF (i.e., a CF male/female ratio of 1.11). This reflects a higher rate of CF female mortality, which is documented in cumulative sex-specific survival rates in a series of 5-year birth cohorts of CF patients from 1976 to 1995 (http://www.cysticfibrosis.ca/assets/files/pdf/cpdr_reporte.pdf). To date, no clear differences have been evident in Canadian CF sex-specific survival rates for birth cohorts from 1996 to 2010, which have as yet largely not (or only briefly) been exposed to post-pubertal levels of sex hormones, including estrogen. The Irish CF data registry (http://www.cfri.ie/docs/annual_reports/CFRI2011.pdf) documents a female disadvantage in sex-specific survival of 5-year birth cohorts going back to the 1980s. A male majority in current populations of CF patients is recorded in annual reports of CF patient registries in the United States, Europe and Australia (Table 1). Consistent with the North American and Australian findings, overall 52.2% of European people with CF in 2009 are male (CF male/female ratio of 1.09; *European Cystic Fibrosis Society Patient Registry, Annual data report, year 2008–2009, version 03.2012*), a proportion that rises to 53.4% (CF male/female ratio of 1.14) among adults >18 years old (http://www.ecfs.eu/files/webfm/webfiles/File/ecfs_registry/ECFSPR_Report0809_v32012.pdf).

Whereas the male/female ratio is higher amongst CF adults than for CF patients of all ages in multiple countries

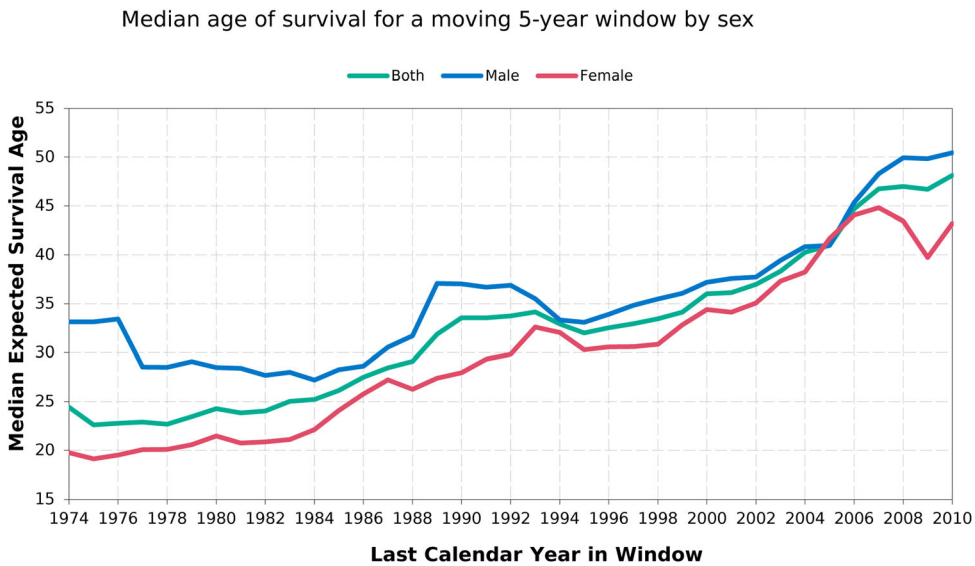

Fig. 1. Reproduced, with permission, from *Cystic Fibrosis Canada CF Registry 2010 Annual Report*; Cystic Fibrosis Canada, Toronto, 2012.

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TABLE 1— Gender Distribution of Persons With Cystic Fibrosis in Patient Registries

Registry	Year	Males % (range)	Dates	URL/citation
Australia	2011	52.3	2011	http://www.cysticfibrosis.org.au/media/wysiwyg/CF-Australia/medical-documents/ACFDR_2011/ACFDR_2011_Report.pdf Australian Cystic Fibrosis Data Registry, 2011 Annual Report; Cystic Fibrosis Australia, Baulkham Hills, NSW, 2012
Canada	2011	52.7 (52.7–54)	1986–2011	http://www.cysticfibrosis.ca/assets/files/pdf/Registry2011FINALOnlineEN.pdf Cystic Fibrosis Canada, Canadian CF Registry, 2011 Annual Report; Toronto, 2013
EuroCFSoc	2009	52.2 (52.2–52.7)	2003–2009	http://www.ecfs.eu/files/webfm/webfiles/File/ecfs_registry/ECFSPR_Report0809_v32012.pdf European Cystic Fibrosis Society Patient Registry, 2008–2009 Annual Data Report, Karup, Denmark, 2012
France	2010	51.4 (51.4–52.4)	2007–2010	http://www.vaincrelamuco.org/e_upload/pdf/rapport_registre_2010.pdf French CF Registry, Annual Data Report 2010; Vaincre la Mucoviscidose and INED. Paris, 2012
Ireland	2011	57.0 (57.0–57.6)	2009–2011	http://www.cfri.ie/docs/annual_reports/CFRI2011.pdf Cystic Fibrosis Registry of Ireland, Annual Report 2011; Dublin, 2012
UK	2011	53.2 (53.1–53.9)	2007–2011	https://www.cysticfibrosis.org.uk/media/82506/CR_Annual_Data_Report_2011_Jan_13.pdf UK CF Registry, Annual Data Report 2011; Cystic Fibrosis Trust, Bromley, Kent, 2013
USA	2011	51.8 (51.8–52.8)	2001–2011	http://www.cff.org/UploadedFiles/research/ClinicalResearch/2011-Patient-Registry.pdf Cystic Fibrosis Foundation Patient Registry, 2011 Annual Data Report. Bethesda, Maryland, 2012

"Year" = year to which the data pertain. "Dates" = the years for which data from this registry report are available. "Males %" = % of all CF patients of all ages that are male. "Range" = minimum and maximum values of "Males %" reported during the "Dates".

(including the UK, France, Ireland and Canada), the male/female ratio in the corresponding general populations drops with increasing age. For example, the CF male/female ratio in France is 1.057 for all ages, but increases to 1.104 for those over 18 years old. By contrast, the estimated 2013 male/female ratio in France for all ages is 0.96, down from a ratio of 1.05 at birth (World Factbook, https://www.cia.gov/library/publications/the-world-factbook/). The corresponding CF data for Ireland are 1.33 for all ages and 1.38 for over 18 years old, and for the UK 1.13 for all ages and 1.23 over 16 years old, whilst the general population trends for these countries are very similar to those of France, USA, Germany, and Canada (World Factbook, https://www.cia.gov/library/publications/the-world-factbook/). Thus there is reasonable evidence for a female disadvantage in survival amongst CF patients that contrasts with a female survival advantage in the general population.

There are national and regional differences with respect to the effects of gender on CF survival, which may reflect differences in the completeness of the data captured from each country. Using data collected on 6,835 patients in 1995–2005 for patients in all age groups, the German Cystic Fibrosis Quality Assessment project did not detect any gender gap in mortality.[7] Nevertheless, the German Registry annual report for the year 2010 (http://muko.info/fileadmin/redaktion/Was_wir_tun/Therapiefoerderung/Mukoviszidose2010_verlinkt.pdf) reported a CF male/female ratio of 1.08 (52% male) including all age groups, as opposed to a corresponding ratio of 0.97 (49% male) in the total population, again suggesting a higher cumulative rate of CF female mortality. The effect of gender on survival of 2,293 children in the Italian Registry for CF who were born after January 1, 1988, was studied longitudinally. Although in this population CF females did not experience a higher mortality than males, due to the disease they lost the expected female survival advantage that is present in the general population at this age.[8]

As treatment for CF has increased in scope and intensity over the years, with progressive improvements in overall survival, some observers have questioned whether the previously noted gap is still found among patients who have benefited from modern clinical care. The existence of the gender gap was disputed by a 2005 report from an individual British pediatric CF clinic,[9] in which two retrospective cross-sectional studies (105 patients in 1993 and 209 patients in 2002) and two 5-year longitudinal studies, beginning in 1993 (40 patients) and in 1998 (81 patients) failed to find any gender—based differences in mortality, pulmonary function (FEV_1) or nutritional status (body mass index, BMI). However, a retrospective review of the period 2000–2009 assessed pooled data from the same institution and another large pediatric CF specialist centre in London, UK. Out of 1,022 children, the numbers of patients who died or

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received transplanted lungs were quite small, 11 deaths and 8 children who had a lung transplant, making the estimates of a female preponderance (82% deaths and 75% transplants) imprecise, although compatible with an ongoing female disadvantage in CF.[10] The ages of death or transplant were comparable between the genders.

In accordance with these more recent data, analysis of 7,937 CF patients from the UK in the year 2010 concluded there is evidence of a more severe lung function deficit in teenage females and that the reduced proportion of females in the adult cohorts is consistent with a survival difference.[11] Although the differences are statistically significant, based on these estimates, the true difference in FEV_1 between males and females could be as high as 10 percentage points (likely of clinical significance) or as low as 1.4 percentage points (not clinically significant). A retrospective cohort analysis of survival in 32,766 CF patients in the United States *Cystic Fibrosis Foundation Patient Registry* between January 1, 1995 and December 31, 2007 also described persistent gender inequality for life expectancy in CF even after the advent of new pulmonary therapies.[12] Kaplan–Meier analysis showed a significant difference between CF men and women ($P < 0.001$) with a median life expectancy of 36.0 years (95% confidence interval [CI] 35.0–37.3) in women and 38.7 years (95% CI 37.8–39.6) in men. CF women had a decreased life expectancy relative to men even after controlling for morphometric factors, sputum microbiology results, and CF related co-morbidities.[12] Therefore, recent data sets support the hypothesis that there is indeed still a gender gap in mortality even among CF patients who have benefited from modern clinical care.

GENDER AND EXOGENOUS SEX HORMONES—EFFECTS ON DISEASE PROGRESSION

Gender affects the progression of CF disease with respect to lung infection (see below), decline in pulmonary function and nutritional status, and the onset of CF Related Diabetes. These interrelated factors in turn have a negative impact on survival. Female gender is an important independent risk factor for rate of decline in pulmonary function in certain age groups.[13,14] Corey et al.[13] reported that at age 5 years males and females did not show different intercepts of lung function, but that lung function decline was more pronounced in females subsequently. By contrast, Konstan et al.[14] reported female gender to be a risk factor for lung function decline in some, but not all, age categories. Therefore, the link between sex hormones and lung disease progression starting at puberty has not been clearly demonstrated in all data sets. In Canada in 2010, 31% of female adults with CF and 19% of male adults with CF were classified as underweight (http://www.cysticfibrosis.ca/assets/files/pdf/cpdr_reporte.pdf). A retrospective analysis of 2,372 patients from the 1986–2008 *US CF Foundation Patient Registry* found that female gender was a significant risk factor for the development of CF Related Diabetes, affecting more females (29%) of this cohort than males (19%), $P < 0.001$.[15]

Although estrogen and progesterone are already in use by CF patients in contraceptives, the effects of contraceptive oral hormone pills on CF lung disease are not well understood. A recent study of the contraceptive pill in CF failed to detect any effect of exogenous female sex hormones (not necessarily exclusively estrogen) on annual change in pulmonary function (Forced Expiratory Volume in one second), body mass index or total days of intravenous antibiotic use in intra-patient and in inter-patient analyses.[16]

ESTROGEN—EFFECT ON INFECTION

Bacterial infection is central to the pathogenesis of CF (reviewed in Ref.[1]). *P. aeruginosa* is a major bacterial pathogen in CF lung disease, contributing to and perpetuating lung inflammation. *P. aeruginosa* leads to mucopurulent plugging, bronchiectasis and intermittent acute pulmonary exacerbations.[1] Chronic *P. aeruginosa* within the CF airway is a negative determinant of prognosis[17] and the onset of mucoid *P. aeruginosa* colonization is associated with subsequent lung function decline.[18,19] Chronic *P. aeruginosa* infection of CF airway is associated with the formation of biofilms.[20] A major phenotypic shift in the growth form of the *P. aeruginosa* from planktonic ("free-swimming") to biofilm is causally linked to the onset of resistance to antibiotic therapy.

There is evidence that estrogens worsen CF lung infections. Circulating levels of the estrogen, estradiol, correlate with infective exacerbations among menstruating women with cystic fibrosis and mucoid *P. aeruginosa* increased in density during periods with high estradiol levels.[21] In vitro, estradiol (or its metabolite estriol) directly enhance the growth of *P. aeruginosa* and the development of a mucoid phenotype.[21] Immune defenses against infection may be classified into *adaptive* (responding to specific pathogens) and *innate* (naturally present and active even upon the first exposure to a pathogen). Antimicrobial peptides (AMPs) are an important part of the innate immune system's capacity to fight infection and inhibit growth of *P. aeruginosa*.[22–25] The importance of AMPs in CF is suggested by the finding that the AMPs lactoferrin, statherin and prolactin-induced protein are more highly expressed in the airway epithelia of CF patients with the best lung function.[26] Estrogen can modulate the formation of *P. aeruginosa* biofilms through its suppression of AMPs, such as lactoferrin, which prevent bacterial biofilm development.[27] In CF male mice exposed to exogenous

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estrogen and then infected with *P. aeruginosa*, there was a virtual elimination of mRNA for lactoferrin in lung tissue and bacterial burden was increased in both lung tissue homogenate and BAL fluid.[28] Suppression of AMPs by estrogen would be expected to aggravate inflammation as AMPs can also inhibit inflammation directly.[22]

ESTROGEN—EFFECTS ON INFLAMMATION

Chronic, dysregulated airway inflammation is a hallmark of CF lung disease, with an exuberant neutrophilic response that is ineffective in reducing *P. aeruginosa* infection but is detrimental, leading to lung tissue destruction. CF patients with *P. aeruginosa* infection have dysregulated cytokine profiles, with elevated levels compared to normal of pro-inflammatory cytokines, including TNFα and interleukin (IL)-8.[29–31] Thus factors that act to increase inflammation in the CF lung should be expected to be harmful, whereas an agent that reduces inflammation should be beneficial.

The increased neutrophils that are found in bronchoalveolar lavage (BAL) fluid[32–34] are the main inflammatory cells in the CF airway lumen. However, in the epithelial submucosa are Th17 helper lymphocytes that produce pro-inflammatory products of the IL-23/IL-17 pathway which appear to be important because they regulate neutrophil recruitment and lung inflammation in CF mice and possibly also in humans. Helper T-cells were long thought to belong exclusively to either of the canonical Th1 or Th2 lineages. However, it is now recognized that the distinct lineage named Th17 is induced by IL-23 to produce IL-17 and other pro-inflammatory Th17 effector molecules. The development, regulation and functional capacities of Th17 cells have been reviewed.[35] Tan et al. point out that the primary function of Th17 cells appears to be the clearance of pathogens that are not adequately handled by Th1 and Th2 cells, in particular extracellular bacteria and fungi. Although clearance of some pathogens depends on this response, certain key CF pathogens, such as *P. aeruginosa* and *Aspergillus fumigatus* induce the production of IL-17 and a strong neutrophilic response but still fail to be cleared. In such situations IL-17-driven inflammation is not protective and may be detrimental.[36] Human CF patients whose lungs are actively infected with *P. aeruginosa* have elevated sputum levels of IL-23 and of IL-17, levels that reduce dramatically following antibiotic treatment.[37,38] Studies of adult male mice demonstrated a critical role for IL-23 (and the Th17 products it induces) in the pathogenesis of murine lung inflammation and neutrophil recruitment in response to airway infection with *P. aeruginosa*.[38,39] Thus lines of evidence suggest a correlation between levels of IL-23/IL-17 and the severity of lung inflammation in both mice and in human CF lung disease.

Sex hormones affect the inflammatory response.[40,41] Adolescents with CF produce normal amounts and quality of endogenous sex hormones and have the biological ability (steroid receptors, etc.) to respond to them. Hence any immunomodulatory effects of female sex hormones that occur in people who do not have CF probably also occur in CF patients. Indeed, recent assessment of the transcriptome of human native nasal epithelial cells expressing F508del-CFTR detected an up-regulation in CF compared to normal controls of estrogen receptor 1 (ESR1) and a series of ESR1-responsive genes, including CXCR2 (also known as IL-8 receptor β), an important enhancer of inflammation in CF airway.[42]

Estrogen has a complex immunomodulatory effect upon inflammation. Importantly, in addition to having pro-inflammatory effects in certain circumstances, estrogen can also have immunosuppressive effects.[43] The activity of cytokine-secreting cells in vivo correlates with sex hormone levels and fluctuates with menstruation.[44] Estrogen stimulates T-cell-dependent immune responses.[45] Sex hormones also modulate the direction of the maturation of T Helper lymphocytes in vitro.[46] The effects are the same on cells of male as of female origin, suggesting these particular changes are indeed due to the hormones, rather than to the genetic differences between the sexes. Khan et al.[47] found in activated mouse splenocytes that estrogen upregulates the levels of IL-17, which were further enhanced by exposure of activated splenocytes to IL-23, particularly in cells from estrogen-treated mice.

In vitro[47] and animal[28] studies suggest that estrogen augments the pro-inflammatory IL-17 family of cytokines, which have been implicated in CF lung disease. We reported that exogenous estrogen stimulates production of IL-17 pathway agonists, increasing the pro-inflammatory environment of the lungs of male CF mice.[28] Estrogen treatment caused adult male CF mice with pneumonia due to a mucoid *P. aeruginosa* CF clinical isolate to develop more severe manifestations of inflammation in both lung tissue and in bronchial alveolar lavage (BAL) fluid, with increased total white blood cell counts and differential and absolute cell counts of polymorphonuclear leukocytes (neutrophils). Since neutrophils enhance the formation of *P. aeruginosa* biofilms,[48] our finding that estrogen increases neutrophil infiltrates in response to *P. aeruginosa* in CF male mouse lung tissue and BAL fluid suggests estrogen has a detrimental effect upon these lungs. Inflammatory infiltrates and mucin production were increased on histology. Increased lung tissue mRNA levels for IL-23 and IL-17 were accompanied by elevated protein levels of Th17-associated pro-inflammatory mediators in BAL fluid. On the other hand, estrogen suppresses the acute lung inflammatory responses of mice to

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lipopolysaccharide-induced injury through an effect on vascular cell adhesion molecules and on proinflammatory mediators.[49] In addition, Chotirmall et al.[50] reported that estrogen caused a reduced production of the pro-inflammatory mediator (IL-8) in CF bronchial epithelium through inhibition of nuclear factor (NF)-κB and IL-8 gene expression. Interestingly, the authors suggested that estrogenic inhibition of IL-8 release by CF bronchial epithelial cells would result in a hyporesponsive innate immune response in the CF lung during times of high estrogen exposure, which then would predispose to infection and bacterial colonization. Therefore, in the setting of an acute infective pulmonary exacerbation, activation of inflammation may be protective.[50] Although their findings do support the hypothesis that estrogen contributes to the CF gender gap, it remains to be clarified which of estrogen's effects are most harmful to the CF lung. Given evidence in a CF animal model of acute *P. aeruginosa* lung infection and in human disease of pro-inflammatory effects (neutrophilic and Th17) of estrogen, and since the bulk of the evidence suggests that inflammation is deleterious in CF, it is reasonable to propose that overall estrogen aggravates CF airway tissue damage by causing a net increase in lung inflammation as well as infection.

Lessons may be learned from other inflammatory lung diseases, including asthma and autoimmune diseases. The incidence and prevalence of asthma reflects a male predominance before puberty and a female one thereafter.[51,52] Female athletes with mild asthma and regular natural menses experience a deterioration in exercise-induced symptoms, lung function and bronchodilator use that correlates with the highest levels of salivary progesterone.[53] Prolonged (female) hormone replacement therapy in post-menopausal women is associated with a higher incidence of new asthma than in untreated controls or in men.[54] It is possible that the effects of female sex hormones on inflammation in CF may resemble the effects seen in asthma.

Sex hormones modulate autoimmunity, a class of illnesses increasingly associated with increased Th17 activity.[55] Autoimmune diseases show distinct gender-based differences in prevalence, with shifts over a lifespan. Systemic lupus erythematosis (SLE) has a female/male ratio of 2.5:1 before puberty, 10:1 after puberty, and it returns to 2.5:1 after menopause. It has been known for almost 30 years that sex hormones modulate autoimmunity in mouse models of SLE, with androgens suppressing and estrogens accelerating disease.[56] The concept that CF may have in common with autoimmune diseases is that excessive inflammation due to over-stimulation of Th17 activity can be harmful, and that further stimulation by estrogen can aggravate the pathology.

SEX HORMONES—EFFECTS ON ION TRANSPORT

In CF, lack of cAMP-regulated CF transmembrane conductance regulator (CFTR) activity diminishes Cl^- and water secretion from airway epithelial cells, reducing the critical capacity to regulate the volume of the airway surface liquid, predisposing to subsequent mucus plugging and infections. Reduced CFTR activity also leads to dysinhibition of the airway epithelial sodium channel (ENaC), enhancing absorption of sodium and airway surface liquid, further compromising mucociliary clearance and aggravating airway infections (reviewed by Boucher[57]).

Sex hormones may modulate the severity of CF lung disease by altering respiratory transepithelial ion transport. Consistent with this hypothesis, we have previously reported that female sex hormones augment levels of mRNA encoding ENaC protein subunits in rat whole lung and increase measures of ENaC ion transport (amiloride-sensitive short-circuit current) across primary monolayer cultures of rat alveolar epithelial cells.[58] We also showed that sex hormones alter CFTR expression and ion transport activity in rabbit and human pancreatic epithelial cells[59] and in rat airway epithelial cells.[60] Subsequently, in studies of young CF women with regular menstrual cycles, we showed that the amiloride-insensitive (non-ENaC) component of the nasal transepithelial electrical potential difference (NPD, a measure of ion transport across the respiratory epithelium) varied with endogenous serum estrogen levels.[61]

Estrogen has a complex series of effects on transepithelial ion transport and the overall net effect of estrogen on CF disease is likely to be the result of an integration of multiple, sometimes opposing, hormonal actions. One estrogenic effect which potentially could have a beneficial effect in CF may involve the up-regulation of Na(+)/H(+)-exchanger regulatory factor 1 (NHERF1), which in turn has been shown to significantly rescue DeltaF508CFTR-dependent chloride secretion in CFBE41o(−) cell monolayers.[62]

CF airways retain a limited capability to regulate airway surface liquid volume through purinergic pathways that raise intracellular Ca^{2+} concentration to stimulate an alternate pathway to Cl^- secretion.[63] Coakley et al.[64] measured NPD to show that UTP-mediated Cl^- secretion was reduced during the periovulatory estrogen maxima in both women with CF and normal, healthy women. Estrogen also inhibited Ca^{2+} signaling and airway surface liquid volume homeostasis in non-CF and CF airway epithelia by attenuating Ca^{2+} influx. This inhibition of Ca^{2+} signaling was prevented and even potentiated by the estrogen antagonist tamoxifen, suggesting that antiestrogens may be beneficial in the treatment of CF lung disease because they increase Cl^- secretion in the airways.[64]

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CONCLUSIONS

Evidence for a CF gender gap in mortality/survival has been described for almost 40 years, and the preponderance of current data indicate that it still exists in the modern era of aggressive treatment of CF lung disease. It will be of interest to explore in future studies if the gender gap will be observed in the recently born cohorts of CF patients who will have lived exclusively in the era of newborn screening and more aggressive treatment, and therefore will have benefited from such therapy all of their lives. The female sex hormone estrogen has been implicated as a factor that modulates crucial components of CF lung physiology including airway infection, dysregulated inflammation and transepithelial ion and fluid transport. Since the bulk of the evidence suggests that inflammation is deleterious in CF, it is reasonable to propose that estrogen aggravates CF airway tissue damage by causing a net increase in lung inflammation. This appears to involve estrogen increasing pro-inflammatory mediators such as the IL-23/IL-17 pathway, while inhibiting AMPs and transepithelial ion and fluid transport, rendering the female CF airway less capable in fighting key CF pathogens. Elucidation of the pathophysiology of hormonal aggravation of CF lung disease may pave the way for novel therapeutic interventions. These could potentially involve manipulation of the airway hormonal milieu, the response of the CF airway to hormonal stimulation, or direct modulation of novel pathological mechanisms identified in the process. Novel therapies for CF, such as estrogen antagonists that are currently used for other indications, could be explored in the future. Delivery of steroid hormone antagonists by inhalation might have the advantage of targeted delivery to the location of greatest need along with reduction of risk of adverse effects by minimization of systemic absorption, analogous to the use of inhaled corticosteroids to suppress airway inflammation in asthma and in CF. Any attempt at modulation of the hormonal milieu would need to be done after careful consideration of the overall effect of estrogens on the pathophysiology of CF, and which of the various individual effects of estrogen on CF have the greatest impact on CF outcome. The magnitude of the gender gap in CF mortality combined with the promise of recent findings strongly suggest that further work in this field is well justified.

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